Engineering Applications of Artificial Intelligence 175 (2026) 114690

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Research paper

An accurate, generalized, and efficient detection framework for steel surface defect inspection

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ARTICLE INFO

Keywords:
Steel surface defect detection
Artificial intelligence–based visual inspection
One-stage object detection framework
Industrial manufacturing quality inspection

ABSTRACT

Steel surface defect detection is crucial for ensuring product quality and process reliability in industrial manufacturing; however, existing methods often struggle to simultaneously achieve high accuracy, real-time efficiency, and strong cross-dataset generalization. This paper proposes AGE-YOLO (Accurate, Generalized, and Efficient YOLO (You Only Look Once)), an enhanced detection framework built upon the YOLO version 11 (YOLOv11) architecture. It is designed to address these challenges by incorporating three key components: (1) a Feature-Context Multi-Aggregation Network (FC-MANet) that integrates Convolutional Gated Linear Units (CGLU) and Partial Convolution (PConv) to capture irregular defect features while mitigating background artifacts through dynamic gating; (2) an Adaptive Cross-scale Fusion (ACF) module that dynamically aligns and fuses multi-level features via joint spatial–channel guidance, effectively suppressing irrelevant background responses; and (3) a Shared Insight Focus Head (SIFHead) employing lightweight shared convolutions with Group Normalization (GN) to enhance localization consistency, particularly for small defects.

Experimental results on the GC10-DET dataset demonstrate that AGE-YOLO achieves a mean Average Precision (mAP) of 72.8% at an Intersection over Union (IoU) threshold of 0.5 (mAP@0.5), outperforming YOLOv11n (nano) baseline by 5.2% and recent YOLOv12n and YOLOv13n variants by 5.8% and 5.6%, respectively. The method also surpasses the Real-Time DEtection TRansformer (RT-DETR) by 3.6% mAP while using only 10.4% of its parameters. With 3.41 million parameters and 7.7 Giga Floating-Point Operations (GFLOPs), AGE-YOLO achieves a real-time inference speed of 189 frames per second (FPS) at 640 × 640 resolution on an NVIDIA RTX 4090D Graphics Processing Unit (GPU). Furthermore, hardware validation on the NVIDIA Jetson Orin Nano platform shows that, after optimization with the NVIDIA TensorRT inference engine using 16-bit floating-point (FP16) precision, the model attains a throughput of 74.6 FPS, confirming its suitability for real-time edge deployment. Additional experiments on the Northeastern University steel defect dataset (NEU-DET) and the printed circuit board defect dataset released by the Peking University Open Laboratory on Human–Robot Interaction (HRIPCB) demonstrate mAP@0.5 scores of 85.4% and 95.7%, respectively, yielding an average improvement of 4.5% over representative lightweight detectors. The implementation of this work is publicly available at https://github.com/gnmtdt/AGE-YOLO.

1. Introduction

Steel manufacturing is a cornerstone of modern industrial development. As a fundamental material widely used in aerospace, automotive, construction, and energy sectors, the quality of steel products directly influences structural performance, production efficiency, and operational safety. With the rapid evolution of automated production lines and intelligent manufacturing systems, industrial requirements for both product quality and throughput have become increasingly stringent. Among various quality indicators, surface integrity is particularly critical because surface defects can significantly degrade mechanical properties and compromise the reliability of downstream components. Consequently, the development of accurate and efficient steel surface defect detection technologies has become essential for ensuring product reliability and advancing industrial modernization.

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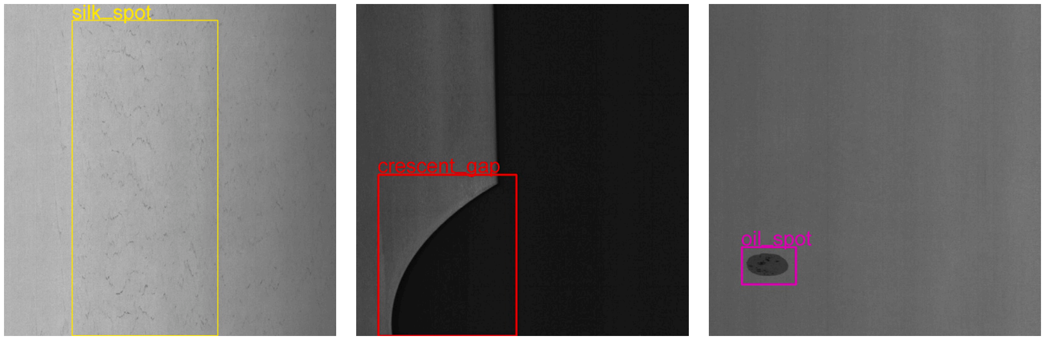
https://doi.org/10.1016/j.engappai.2026.114690
Received 25 November 2025; Received in revised form 28 February 2026; Accepted 31 March 2026
Available online 2 April 2026



(a) Unstructured defect (b) Large-scale defect (c) Small-scale defect

Fig. 1. Representative samples from the GC10-DET dataset illustrating the diversity of defect characteristics: (a) a low-contrast, unstructured silk spot; (b) a large-scale crescent gap with pronounced geometric variation; and (c) a small-scale oil spot requiring precise localization.

Existing inspection approaches can generally be categorized into manual inspection, traditional machine vision, and deep learning-based methods. Manual inspection relies heavily on human expertise, which makes it labor-intensive, subjective, and unsuitable for large-scale production environments. Traditional machine vision techniques utilize handcrafted features, such as texture descriptors, edge information, and intensity gradients, providing a certain degree of automation but often lacking robustness under complex and variable industrial conditions. In contrast, deep learning approaches, particularly convolutional neural networks (CNNs), have demonstrated superior performance by learning hierarchical feature representations directly from data. These methods have become the dominant paradigm for industrial defect detection, offering improved robustness, higher accuracy, and real-time capability compared with conventional techniques (Liu et al., 2023).

Despite these advances, the intrinsic complexity and variability of steel surface conditions continue to pose significant challenges for automated defect detection, as illustrated in Fig. 1.

First, many industrial defects, such as silk spots, exhibit unstructured and irregular morphologies with low contrast, as illustrated in Fig. 1(a). These characteristics make them difficult to distinguish from complex industrial backgrounds, thereby requiring advanced nonlinear feature extraction mechanisms to effectively model non-uniform textures. Second, defects often present significant scale variability and are frequently embedded in noisy environments. Large-scale defects, such as the crescent gap shown in Fig. 1(b), require robust cross-scale feature alignment to integrate multi-level semantic information while suppressing irrelevant background responses. Third, the presence of minute defects, exemplified by the oil spot in Fig. 1(c), poses substantial challenges for localization consistency. Reliable detection of such small targets demands high sensitivity and stable prediction heads, particularly under the computational constraints typical of real-time edge deployment.

Beyond these intrinsic morphological challenges, the operational reliability of surface inspection systems is further limited by harsh production environments. In practical rolling metal manufacturing, inspection systems are frequently exposed to inhomogeneous illumination and mechanical vibrations (Maruschak et al., 2025), which are characteristic of real industrial settings. Recent studies indicate that fluctuations in illuminance and low-amplitude vibrations can significantly degrade the performance of CNN-based defect recognition systems, highlighting persistent barriers to reliable deployment in real-world scenarios.

Extensive research has further advanced deep learning-based industrial inspection. H. Li et al. (2025) proposed the Re-parameterized Bilateral Path Aggregation Network (RepBi-PAN), a hierarchical feature integration framework that combines a Densely Connected Network (DenseNet) with a normalization-based attention mechanism to enhance robustness in complex backgrounds. Zheng et al. (2025) integrated Legendre multiwavelets with a Feature Attention Guidance (FAG) mechanism to construct a lightweight yet high-accuracy network capable of capturing complex geometric patterns across multiple scales. Bai et al. (2024) presented a comprehensive survey of surface defect detection research in the 2020s, emphasizing systemic challenges in smart manufacturing and the need for models with stronger adaptability to diverse industrial conditions. Meanwhile, Zhang et al. (2025) developed an improved YOLOv8 framework incorporating an implicit prediction head and the Inner-Wise Intersection over Union (Inner-WIoU) loss, achieving enhanced localization accuracy and robustness.

Despite these advances, most existing approaches still struggle to reconcile benchmark performance with real-world industrial constraints. Specifically, three critical research gaps remain:

- **Inadequate morphological modeling**: Current convolutional kernels lack the dynamic selection capability required to represent unstructured textures, often leading to high false-positive rates when local background artifacts mimic the low-contrast features of defects.
- **Suboptimal cross-scale alignment**: Existing fusion architectures frequently fail to maintain semantic consistency during multi-level integration, which results in information attenuation and global background interference during the feature aggregation process.
- **Efficiency-precision bottleneck**: Achieving high localization precision for small targets typically requires heavy parameterization, creating a computational redundancy that hinders real-time deployment on resource-constrained edge platforms.

To address these identified gaps, this study proposes Accurate, Generalized, and Efficient You Only Look Once (AGE-YOLO), an enhanced framework based on the YOLOv11 architecture. The main contributions are as follows:

- **Feature extraction enhancement**: The Feature-Context Multi-Aggregation Network (FC-MANet) utilizes Convolutional Gated Linear Unit (CGLU)-based gating and Partial Convolution (PConv) to achieve adaptive feature selection. This mechanism suppresses local background artifacts by filtering out non-defect textural responses, improving the modeling of unstructured morphologies.
- **Feature fusion optimization**: The Adaptive Cross-scale Fusion (ACF) module introduces dual spatial–channel guidance to optimize feature alignment. This ensures cross-scale semantic consistency while purifying global background noise during the fusion process.
- **Detection head refinement**: The Shared Insight Focus Head (SIF-Head) adopts lightweight shared convolutions and Group Normalization (GN) to improve localization stability with reduced computational overhead, specifically enhancing sensitivity to fine-grained small-scale targets.

Table 1
Comparison of AGE-YOLO and representative industrial defect detection paradigms across key technical dimensions.

Methodology	Unstructured perception	Cross-scale alignment	Edge-side efficiency
Traditional CV (Liu et al., 2023)	Low	Very low	High
Standard YOLOv11 (Khanam and Hussain, 2024)	Medium	Medium	High
Specialized SOTA (H. Li et al., 2025; Xie et al., 2025)	Medium	High	Medium
AGE-YOLO (Ours)	**High**	**High**	**High**

To emphasize the distinctive characteristics of the proposed AGE-YOLO, a qualitative comparison with representative industrial detection paradigms is provided in Table 1. The assessment is based on the experimental observations and conclusions reported in the corresponding literature.

Together, these modules establish a unified and lightweight detection architecture that jointly improves detection accuracy and inference efficiency, while maintaining consistent performance across different industrial defect datasets.

2. Related work

Existing research on object detection for industrial surface inspection has primarily focused on four critical architectural components: (1) real-time detection frameworks, (2) feature extraction backbones, (3) multi-level feature fusion strategies, and (4) specialized detection head designs. A concise review of how these components have evolved to meet industrial constraints is presented below.

2.1. Real-time object detectors

With the continuous advancement of computing hardware and deep learning algorithms, object detection has achieved remarkable progress, particularly in industrial surface defect analysis. Existing detection frameworks can be categorized into one-stage, two-stage, and Transformer-based paradigms. One-stage detectors integrate localization and classification within a unified network, achieving a favorable balance between accuracy and speed, while two-stage frameworks pursue higher precision through region proposal mechanisms at the expense of real-time efficiency. Transformer-based architectures, such as DEtection TRansformer (DETR), leverage global self-attention mechanisms for contextual reasoning but typically require higher computational resources, which may constrain deployment in latency-sensitive industrial environments.

In recent years, the You Only Look Once (YOLO) family has become the dominant one-stage framework due to its simplicity and efficiency. The evolution from YOLOv3 (Redmon and Farhadi, 2018) to YOLOv8 (Jocher et al., 2023), and the latest iterations from YOLOv11 (Khanam and Hussain, 2024) to YOLOv13 (Lei et al., 2025), reflects steady progress in real-time object detection. Earlier versions such as YOLOv3 introduced multi-scale detection heads for spatial representation, while YOLOv5 and YOLOv8 emphasized modular design and anchor-free prediction to improve flexibility and generalization. More recently, the YOLOv11–v13 series has continued to refine the balance between detection accuracy and computational efficiency. Compared with later variants that introduce more complex attention-driven components, YOLOv11 maintains a streamlined structural design, making it suitable for targeted architectural customization in industrial applications.

Despite these advancements, existing detectors — including recent general-purpose variants — still exhibit limitations when handling unstructured textures, significant scale variations, and small-scale defect instances in steel inspection scenarios. These models often fail to capture low-contrast textural anomalies, lose semantic consistency across scales, or suffer from latency on resource-constrained edge devices. To bridge these research gaps, the proposed AGE-YOLO builds upon the mature architecture of YOLOv11. It introduces an enhanced framework that simultaneously improves accuracy, adaptability, and inference efficiency through three specialized modules: the FC-MANet for discriminative morphological representation, the ACF for dynamic cross-scale alignment, and the SIFHead for consistent small-target localization.

2.2. Feature extraction networks

Feature extraction networks are the cornerstone of modern vision tasks. Visual Geometry Group (VGG) network (Simonyan and Zisserman, 2015) first demonstrated the benefits of depth, while Residual Network (ResNet) (He et al., 2016) introduced skip connections to alleviate vanishing gradients. Later, Res2Net (Gao et al., 2019) and ResNeXt (Xie et al., 2017) improved multi-scale representation via hierarchical and grouped convolutions. In industrial inspection, recent works have tailored these designs for defect detection. For example, Sui and Wang (2025) proposed AMMENet with parallel residual attention to enhance aluminum surface feature extraction, while Feng et al. (2024) developed Hyper-YOLO using a multi-branch MANet module to strengthen perception of unstructured textures. In YOLOv11, the C3K2 module introduces split-convolution blocks for efficient multi-scale encoding. Nevertheless, lightweight extractors often lack sufficient adaptive capacity to model irregular and non-uniform defect morphologies. The proposed FC-MANet introduces dynamic nonlinear filtering to enhance discriminative feature selection under constrained computational budgets.

2.3. Multi-scale feature representation

Accurate detection of objects with varying sizes requires effective multi-scale representation. Specifically, shallow layers capture fine spatial details, while deeper layers encode abstract semantics (Sun et al., 2019). The Feature Pyramid Network (FPN) (T.-Y. Lin et al., 2017) introduced top-down fusion with lateral connections, and the Path Aggregation Network (PAN) (Liu et al., 2018) added a bottom-up path for richer context. Subsequent works further optimized cross-scale fusion: Su et al. (2024) combined salient and background cues to reduce interference, Peng et al. (2024) linked encoder–decoder stages to strengthen multi-scale aggregation, and Chen et al. (2021) mitigated spatial precision loss via parallel bi-fusion paths. YOLOv11 refines feature interaction across scales. However, many existing fusion strategies rely on fixed aggregation patterns, limiting their adaptability in complex industrial backgrounds. The proposed ACF module adopts a dual-guided mechanism to dynamically coordinate spatial and channel interactions, thereby improving cross-scale consistency.

2.4. Detection head

The detection head directly predicts object categories and bounding boxes within a single-stage pipeline (Zou et al., 2023). To enhance localization stability, architectures like FCOS (Tian et al., 2020) pioneered the use of Group Normalization (GN) in anchor-free heads, while TOOD (Feng et al., 2021) focused on aligning classification and localization tasks. For industrial scenarios, recent efforts have prioritized hardware-friendly designs. Z. Li et al. (2025) extended YOLOv8n with spatial–channel attention, and Huang et al. (2025) employed shared convolutions to mitigate parameter redundancy. YOLOv11 currently balances these needs using depthwise separable convolutions. In practice, lightweight detection heads may exhibit feature calibration inconsistency and reduced localization stability. The proposed SIFHead mitigates these issues through shared convolutional representations and scale-aware normalization, strengthening prediction reliability without increasing computational overhead.

3. Methods

3.1. Improved model overview

YOLOv11 (Khanam and Hussain, 2024), developed by the Ultralytics team, is a representative one-stage object detector that follows a standard three-stage pipeline, consisting of a backbone, a feature aggregation network, and a detection head. The backbone progressively reduces spatial resolution while expanding channel capacity through a series of convolutional and attention-based blocks, including Conv, C3k2, SPPF, and C2PSA. This hierarchical design enables the extraction of multi-scale feature representations that jointly encode local structural details and high-level semantic information.

To integrate features across different resolutions, YOLOv11 employs a Feature Pyramid Network (FPN) (T.-Y. Lin et al., 2017) combined with a Path Aggregation Network (PAN) (Liu et al., 2018). The FPN propagates high-level semantic cues to higher-resolution layers via a top-down pathway, which benefits coarse object recognition, while the PAN enhances spatial localization through bottom-up feature reinforcement. Based on these aggregated features, the detection head adopts three parallel branches corresponding to large-, medium-, and small-scale feature maps, enabling scale-aware prediction across diverse object sizes. In addition, YOLOv11 incorporates lightweight design strategies, such as depthwise separable convolutions and partial parameter sharing, to balance detection accuracy and computational efficiency.

Despite its favorable trade-off between performance and efficiency, YOLOv11 still encounters challenges when applied to industrial surface defect inspection. Specifically, small-sized defects, irregular geometric patterns, and low-frequency defect features are prone to information attenuation during deep feature extraction and multi-scale fusion. These limitations motivate recent efforts to enhance YOLO-based architectures with improved feature modeling and cross-scale interaction mechanisms.

Recent studies have validated the effectiveness of specialized architectural modifications. For instance, Xie et al. (2025) demonstrated that re-parameterized feature aggregation can significantly enhance mAP under constrained budgets, while Gao et al. (2025) leveraged Mamba-based modules to strengthen cross-scale consistency. Drawing upon these insights, we propose AGE-YOLO, which transitions from general-purpose detection to a domain-specific architecture tailored for the morphological and scale complexities of steel surface defects.

An architectural comparison between the baseline YOLOv11 and the proposed AGE-YOLO is presented in Fig. 2. As illustrated in Fig. 2(a), the original YOLOv11 adopts a conventional backbone–neck–head paradigm with static FPN+PAN feature aggregation. In contrast, AGE-YOLO (Fig. 2(b)) incorporates three targeted structural enhancements to address the identified research gaps.

First, the FC-MANet modules (highlighted in red) is embedded within both the backbone and neck stages. By leveraging CGLU-based gating and PConv, this module enhances nonlinear representation capacity and adaptive feature selection, enabling more effective modeling of unstructured and irregular defect patterns. Second, the feature fusion pathway is refined through the ACF module (shown in blue), which introduces joint spatial–channel guidance to improve cross-scale semantic alignment while suppressing background interference. Third, the SIFHead (indicated in green) replaces the standard detection head. It utilizes lightweight shared convolutions and group-based scale-aware normalization to improve localization stability and sensitivity to small-scale defects, thereby mitigating the efficiency–precision trade-off for edge deployment.

Overall, AGE-YOLO incorporates three complementary modules, each addressing a specific bottleneck in industrial surface defect detection. By jointly improving feature representation, cross-scale interaction, and detection head efficiency, the proposed framework achieves a balanced enhancement in detection accuracy, computational efficiency, and robustness across diverse defect scenarios.

3.2. FC-MANet module

Industrial surface defects often exhibit irregular geometries, weak contrast, and fragmented spatial structures, which pose challenges for standard convolutional blocks relying on linear feature aggregation. To address these limitations, we propose the Feature–Context Multi-Aggregation Network (FC-MANet) as a lightweight enhancement to the YOLOv11 backbone, aiming to improve nonlinear feature modeling and context-aware representation under complex defect patterns.

FC-MANet replaces the original C3k2 block with a multi-branch structure inspired by MANet (Feng et al., 2024), while incorporating a specialized residual branch composed of Faster-CGLU blocks. This architecture is designed to decompose feature extraction into complementary pathways, enabling the simultaneous preservation of low-level spatial details, local structural cues, and deep nonlinear interactions.

3.2.1. Design motivation and module overview

As illustrated in Fig. 3, FC-MANet adopts a multi-branch aggregation strategy to disentangle different aspects of feature representation. Given an input feature map, the module first performs a linear channel projection to expand the feature space, and subsequently processes the features through multiple parallel pathways, each targeting a specific representation objective.

Formally, given an input feature map $X \in \mathbb{R}^{B\times C_{\text{in}}\times H\times W}$, a 1×1 convolution is applied to obtain a base representation Y with $2C$ channels:

$$Y = \text{Conv}_{1\times 1}(X), \quad Y \in \mathbb{R}^{B\times 2C\times H\times W}. \tag{1}$$

The projected feature map Y is then processed through multiple pathways. The first branch Y_0 and second branch Y_1 are defined as:

$$Y_0 = \text{Conv}_{1\times 1}(Y), \quad Y_1 = \text{Conv}_{1\times 1}(\text{DWConv}_{k\times k}(\text{Conv}_{1\times 1}(Y))). \tag{2}$$

Simultaneously, Y is split into two identity components Y_2 and Y_3:

$$(Y_2, Y_3) = \text{Split}(Y). \tag{3}$$

The component Y_3 serves as the input to a stack of n Faster-CGLU blocks, where each block generates a refined feature map. Let $\mathcal{M}_i$ denote the ith block:

$$Y_3^{(i)} = \mathcal{M}_i(Y_3^{(i-1)}), \quad i = 1,\dots,n, \quad \text{with } Y_3^{(0)} = Y_3. \tag{4}$$

Finally, the outputs from all branches and intermediate blocks are concatenated to form the final multi-aggregated representation:

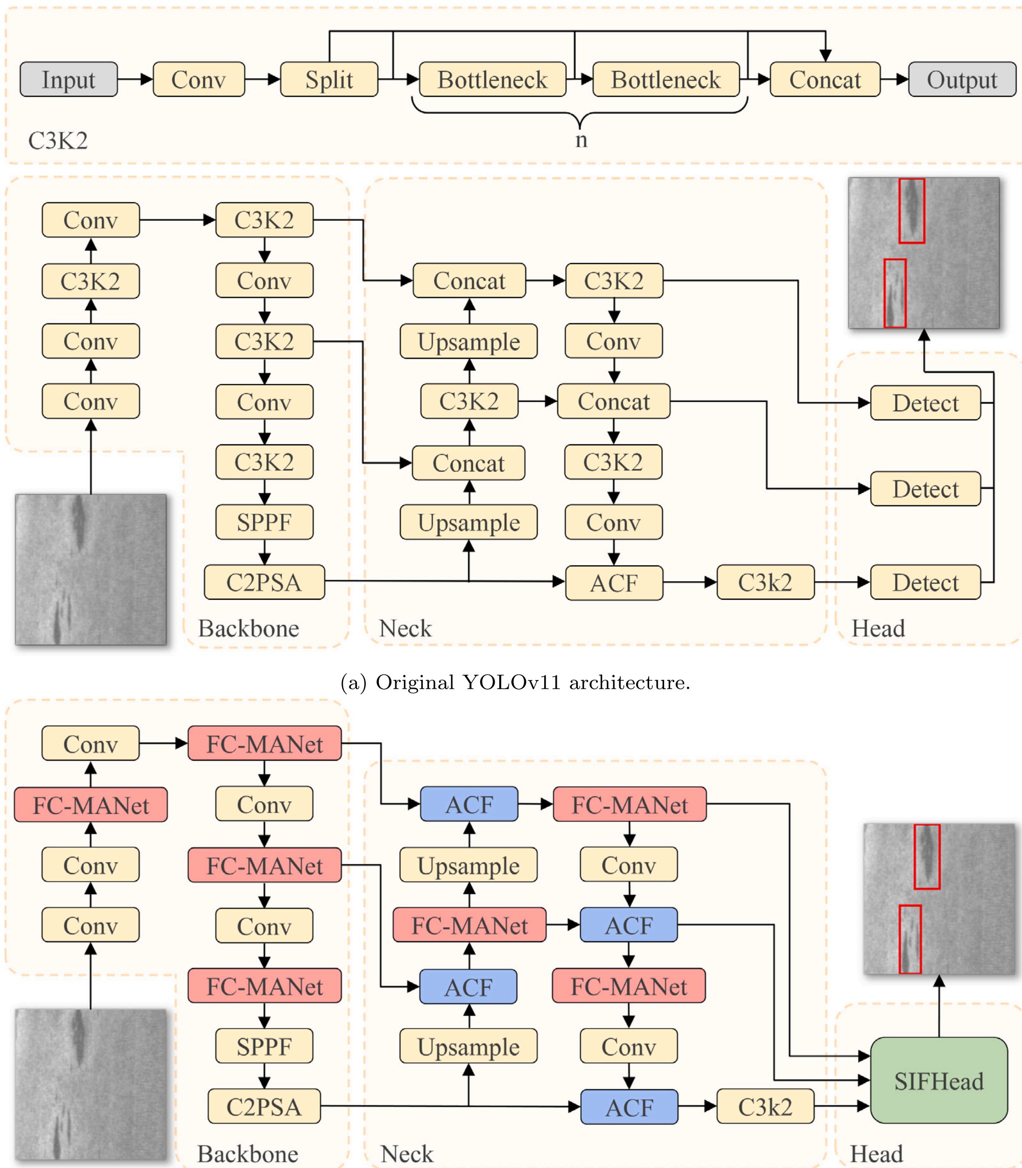
$$F = \text{Conv}_{1\times 1}(\text{Concat}[Y_0, Y_1, Y_2, Y_3, Y_3^{(1)}, \dots, Y_3^{(n)}]). \tag{5}$$

3.2.2. Faster-CGLU: Lightweight gated residual modeling

The Faster-CGLU block, illustrated in the upper-right panel of Fig. 3, serves as the core nonlinear component within FC-MANet. It is designed to enhance channel-wise interaction while maintaining computational efficiency.

Given an input feature X, a spatial-aware Partial Convolution (PConv) is first applied to reduce redundancy and emphasize informative regions:

$$X_{\text{spatial}} = \text{PConv}_{3\times 3}(X). \tag{6}$$



(a) Original YOLOv11 architecture.

(b) Proposed AGE-YOLO architecture.

Fig. 2. Architectural comparison between the baseline and the proposed framework. Relative to YOLOv11 (a), AGE-YOLO (b) integrates three modular innovations: FC-MANet (red) for enhanced feature representation, the ACF module (blue) for adaptive cross-scale fusion, and SIFHead (green) for stable and precise localization. These components are strategically embedded throughout the pipeline to improve the synergy between feature extraction and aggregation.

PConv is employed to exploit channel redundancy commonly observed in industrial surface textures. By operating on only a subset of input channels, PConv reduces memory access cost (MAC) and computational overhead while preserving representational capacity.

The spatially mixed features are subsequently processed by the Convolutional Gated Linear Unit (CGLU). As illustrated in Fig. 3, the input feature map X_{spatial} is first divided into a value branch and a gating branch. The value branch V performs a linear projection to extract primary features, whereas the gating branch G integrates local contextual information through a depthwise convolution to produce a modulation signal:

$$(X_v, X_g) = \text{Split}(X_{\text{spatial}}), \tag{7}$$

$$V = \text{Conv}_{1\times1}(X_v), \quad G = \text{DWConv}_{3\times3}(\text{Conv}_{1\times1}(X_g)). \tag{8}$$

The final output is computed by modulating the value branch with the GELU-activated gating signal, followed by a channel projection and a residual connection to stabilize gradient flow:

$$X_{\text{out}} = X + \text{DropPath}(\text{Conv}_{1\times1}(V \odot \text{GELU}(G))), \tag{9}$$

where $\odot$ represents element-wise multiplication.

From a theoretical perspective, this gated linear mechanism enables element-wise feature modulation, acting as a nonlinear filter that adaptively suppresses low-frequency background noise while amplifying high-frequency defect-sensitive signals. This property is particularly advantageous for detecting low-contrast steel surface defects.

3.2.3. Discussion and advantages

By integrating gated residual modeling into a multi-branch aggregation framework, FC-MANet transitions feature extraction from conventional additive fusion toward a selective information modulation paradigm. This design allows the network to adaptively emphasize defect-relevant features while suppressing background interference, specifically addressing the challenges posed by unstructured and low-contrast steel surface anomalies. Furthermore, by utilizing PConv and

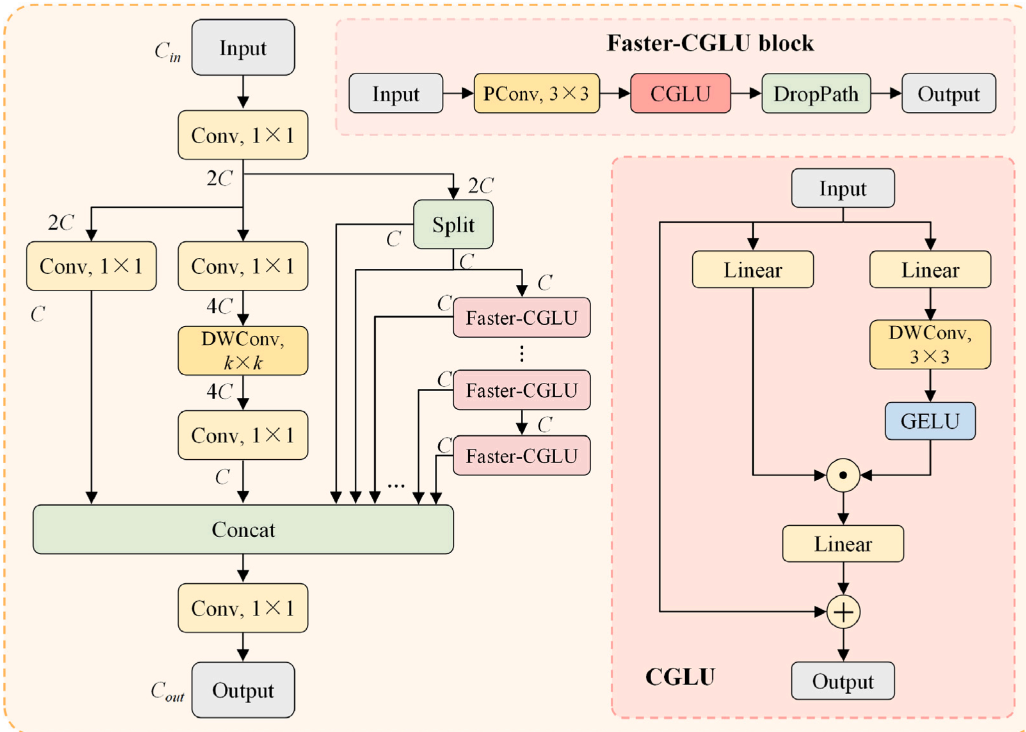


Fig. 3. Structure of the proposed FC-MANet module. The multi-branch architecture integrates local feature extraction and gated nonlinear residual modeling through Faster-CGLU blocks.

CGLU instead of heavy self-attention mechanisms, FC-MANet achieves high discriminative power with significantly lower computational overhead. Consequently, this module provides a more robust and efficient foundation for the subsequent cross-scale feature fusion process.

3.3. ACF module

Accurate detection of objects with varying sizes requires effective multi-scale representation. Although hierarchical backbones naturally provide multi-scale features, conventional fusion strategies such as FPN (T.-Y. Lin et al., 2017) and PAN (Liu et al., 2018) typically rely on fixed summation or concatenation. Such static schemes lack the flexibility to handle the semantic-spatial misalignment inherent in steel surface defects. To address this, we propose the Adaptive Cross-scale Fusion (ACF) module, which replaces rigid integration with a dynamic, dual-attention guided mechanism and explicit cross-level compensation to ensure feature consistency across scales.

3.3.1. Design motivation and module overview

Industrial defects vary significantly in size, texture continuity, and spatial distribution. High-level features encode strong semantic context but lose fine-grained details, while low-level features preserve spatial precision but are susceptible to background noise. Directly fusing these heterogeneous representations often leads to feature dilution. The core motivation of ACF is to selectively recalibrate informative components through dual-attention filtering and to bridge the semantic gap between resolutions using learnable compensation parameters, ensuring that defect-sensitive signals remain prominent during aggregation.

As illustrated in Fig. 4, the ACF module receives two feature maps from different backbone stages, denoted as $x_0 \in \mathbb{R}^{B\times C_1\times H\times W}$ and $x_1 \in \mathbb{R}^{B\times C_2\times H\times W}$. To ensure compatibility during fusion, a 1×1 convolution is first applied to align the channel dimension of x_0 with that of x_1, followed by concatenation along the channel dimension to form a joint representation $X \in \mathbb{R}^{B\times 2C\times H\times W}$. This fused representation is subsequently refined by two complementary attention mechanisms, aiming to adaptively recalibrate channel importance and spatial responses.

$$x_0' = \mathrm{Conv}_{1\times1}(x_0), \quad x_1' = x_1 \tag{10}$$

3.3.2. Dual-attention guided feature refinement

The first attention stage adopts a channel-wise squeeze-and-excitation (SE) (Hu et al., 2018) mechanism to model inter-channel dependencies:

$$X_{se} = \sigma\left(W_2\,\delta\left(W_1\,\mathrm{GAP}(X)\right)\right) \odot X, \tag{11}$$

where $\mathrm{GAP}(\cdot)$ denotes Global Average Pooling, $\delta(\cdot)$ represents the GELU activation function, and $\sigma(\cdot)$ is the sigmoid function. $W_1 \in \mathbb{R}^{\frac{C}{r}\times C}$ and $W_2 \in \mathbb{R}^{C\times\frac{C}{r}}$ are the weights of two fully connected layers with reduction ratio r.

Following channel recalibration, a spatial-aware Position Attention (PA) mechanism is applied to highlight defect-relevant regions:

$$M_s = \sigma\left(\mathrm{Conv}_{7\times7}(X_{se})\right), \quad X_{attn} = M_s \odot X_{se}. \tag{12}$$

The attention-enhanced features are then projected back to the target channel dimension:

$$X_f = \delta\left(\mathrm{Conv}_{1\times1}(X_{attn})\right). \tag{13}$$

This dual-attention design allows ACF to jointly model "what" (channel importance) and "where" (spatial relevance), which is particularly beneficial for detecting small or low-contrast defects embedded in complex backgrounds.

3.3.3. Adaptive cross-scale compensation mechanism

To further balance information exchange between different feature levels, ACF introduces a learnable cross-compensation mechanism, as shown in the right part of Fig. 4. The fused feature X_f is first evenly split along the channel dimension:

$$(X_{0w}, X_{1w}) = \mathrm{Split}(X_f). \tag{14}$$

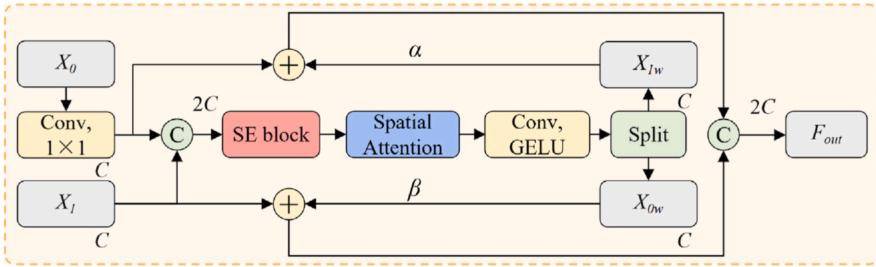


Fig. 4. Architecture of the Adaptive Cross-scale Fusion (ACF) module. It features a dual-attention pipeline for channel-spatial recalibration and a learnable cross-compensation mechanism to dynamically balance semantic and structural information across different feature levels.

Instead of independently propagating these features, ACF employs learnable channel-wise scaling parameters α and β to enable controlled inter-branch enhancement. Based on our implementation, the refined weights from one scale are used to compensate the features of the other:

$$\hat{x}_0 = x'_0 + X_{1w} \cdot \alpha, \quad \hat{x}_1 = x'_1 + X_{0w} \cdot \beta, \tag{15}$$

where $\alpha, \beta \in \mathbb{R}^{1\times C\times 1\times 1}$ are trainable parameters initialized with unit values to ensure training stability.

Finally, the compensated features are concatenated to form the output of the ACF module:

$$F_{\text{out}} = \text{Concat}[\hat{x}_0, \hat{x}_1]. \tag{16}$$

3.3.4. Discussion and advantages

Compared with rigid fusion schemes, ACF introduces a dynamic balancing strategy that explicitly accounts for the complementary nature of multi-scale features. While FC-MANet focuses on local texture-level filtering, ACF operates as a macro-level recalibration mechanism. The dual-attention pipeline functions as a nonlinear spatial–channel filter; it highlights defect-sensitive regions while concurrently suppressing task-irrelevant background interference across the entire feature map. This ensures that prominent background structures do not overwhelm the subtle semantic cues of small defects during feature aggregation.

Overall, ACF offers three tangible advantages: (i) targeted feature refinement via joint channel and spatial attention, (ii) bi-directional alignment of semantic and structural information through learnable compensation parameters α and β, and (iii) a lightweight architecture that maximizes inference throughput.

Within the AGE-YOLO framework, ACF complements the Faster-CGLU-based modulation to establish a multi-stage feature enhancement process. By utilizing efficient convolutional operations instead of computationally expensive self-attention blocks, this design maintains strong representational power without increasing latency. Regarding numerical stability, feature responses are balanced prior to concatenation to maintain consistent variance across scales, preventing gradient dominance by any single resolution. Furthermore, the cross-compensation mechanism employs unit-value initialization for α and β. This strategy establishes an identity-based skip connection from the onset of training, promoting robust gradient propagation and ensuring that multi-scale information is integrated without early-stage attenuation.

3.4. Shared Insight Focus Head (SIFHead)

In industrial surface defect detection, the detection head is responsible for translating multi-scale features into precise localization and classification results. This task is particularly challenging for small-scale defects, where standard detection heads often suffer from feature dilution and localization instability. Conventional YOLO-style heads typically employ scale-specific convolutional branches, which not only introduces significant parameter redundancy but also limits the ability of the model to learn scale-invariant defect representations. To address these issues, we propose the Shared Insight Focus Head (SIFHead), which emphasizes shared representation learning, parameter-free feature refinement, and adaptive scale calibration.

As illustrated in Fig. 5, SIFHead is designed under a "lightweight-yet-shared" philosophy. Instead of maintaining independent prediction branches for each scale, the proposed head enforces parameter sharing across resolutions, encouraging the learning of scale-consistent representations while significantly reducing model complexity.

3.4.1. Channel compression and shared convolutional encoding

As shown in the left part of Fig. 5, multi-scale features F_i from the neck are first projected into a unified channel space with dimension C using 1×1 convolutions:

$$F_i^{\text{gn}} = \text{GN}\big(\text{Conv}_{1\times 1}(F_i)\big). \tag{17}$$

Group Normalization (GN) is adopted instead of Batch Normalization (BN) to ensure batch-size independence. In industrial settings, high-resolution inputs often constrain the feasible batch size, resulting in unstable statistics for BN. GN therefore enables AGE-YOLO to maintain stable performance when transitioning from resource-rich training environments to resource-constrained edge deployment scenarios.

To reduce parameter redundancy while retaining expressive power, SIFHead employs shared depthwise separable convolutions across all feature scales, represented as the green SConv_GN blocks in Fig. 5. Each operation factorizes a standard convolution into a depthwise convolution followed by a pointwise convolution, with weights shared across feature levels:

$$F_i^{\text{shared}} = f_{\text{SConv}}\left(F_i^{\text{gn}}\right), \tag{18}$$

where f_{SConv} denotes the shared separable convolutional transformation.

This design enforces scale-invariant feature encoding, allowing the detection head to capture common defect patterns across resolutions while maintaining a compact parameter footprint.

3.4.2. Attention-based feature refinement

Although shared convolutions enhance cross-scale consistency, they may suppress scale-specific saliency cues. To selectively reinforce defect-relevant responses, SIFHead integrates a parameter-free SimAM attention module (Yang et al., 2021), depicted as the central pink block in Fig. 5. This module computes 3D attention weights for each neuron based on an energy-function formulation:

$$e_n^* = \frac{4(\hat{\sigma}^2 + \lambda)}{(t - \hat{\mu})^2 + 2\hat{\sigma}^2 + 2\lambda} \tag{19}$$

where t and n are the target neuron and other neurons in the same channel, while $\hat{\mu}$ and $\hat{\sigma}^2$ are the local mean and variance, and λ is a hyper-parameter to ensure numerical stability. This analytical approach allows SimAM to recalibrate both spatial and channel-wise responses without introducing additional learnable parameters, making it more expressive than Squeeze-and-Excitation (SE) (Hu et al., 2018) or Convolutional Block Attention Module (CBAM) (Woo et al., 2018) under a

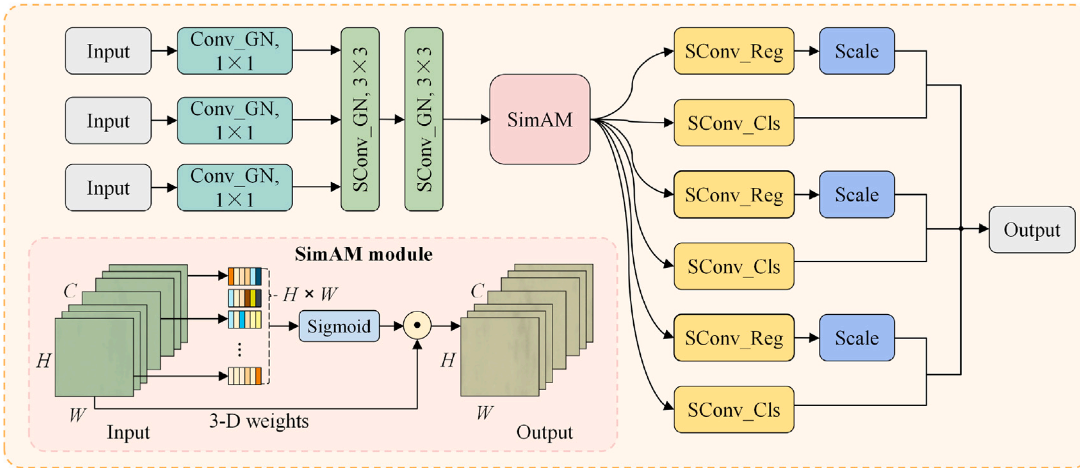


Fig. 5. Architecture of the Shared Insight Focus Head (SIFHead). Multi-scale features are first channel-compressed using 1 × 1 convolutions with group normalization. All scales then share depthwise separable convolutional layers and a SimAM-based attention module for feature enhancement. Finally, shared classification and regression heads generate predictions, with scale-specific compensation layers applied to the regression outputs.

limited parameter budget. The resulting enhanced feature is obtained as:

$$F_i^{\mathrm{enh}} = \mathcal{A}_{\mathrm{SimAM}}\left(F_i^{\mathrm{shared}}\right) \odot F_i^{\mathrm{shared}}, \tag{20}$$

where $\odot$ denotes element-wise modulation of the feature responses.

By highlighting spatially and channel-wise salient activations, this attention mechanism is particularly effective for small or weakly textured defects that are easily overwhelmed by background patterns.

3.4.3. Shared prediction heads with adaptive scale compensation

The enhanced features are finally passed to the prediction stage shown on the right of Fig. 5. Instead of using scale-specific heads, SIFHead employs two globally shared prediction branches: a classification head C_{cls} and a regression head C_{reg}, both implemented using shared separable convolutions (yellow blocks). The regression head follows the Distribution Focal Loss formulation (T. Lin et al., 2017):

$$\hat{P}_i = \sigma\left(C_{\mathrm{cls}}(F_i^{\mathrm{enh}})\right), \quad \hat{B}_i = C_{\mathrm{reg}}(F_i^{\mathrm{enh}}), \tag{21}$$

where $\sigma(\cdot)$ denotes the sigmoid activation.

Although the prediction heads are shared, different feature scales still exhibit distinct regression magnitudes. To compensate for this discrepancy, SIFHead introduces lightweight scale-specific compensation factors, implemented as learnable Scale layers:

$$\tilde{B}_i = s_i \cdot \hat{B}_i, \tag{22}$$

where $i \in \{1, 2, 3\}$ indexes the detection scales and s_i denotes the corresponding scale factor.

This mechanism preserves scale adaptability while retaining the benefits of shared parameterization.

3.4.4. Discussion and advantages

SIFHead transforms the conventional multi-branch detection head into a unified architecture that balances efficiency with localization precision. By utilizing shared separable convolutions and parameter-free SimAM attention, the head enhances sensitivity to fine-grained defect cues while maintaining a compact parameter footprint. Furthermore, the adoption of GN ensures batch-size independence and stable statistical estimation, which is critical for maintaining performance during resource-constrained industrial edge deployment. Combined with scale-specific compensation factors, these design choices collectively enable SIFHead to achieve robust localization and consistent regression for small-scale surface defects, even under constrained computational budgets.

Table 2
Training hyperparameters used in the experiments.

Hyperparameter	Setting
Optimizer	Stochastic Gradient Descent (SGD)
Initial learning rate	0.01
Learning rate schedule	Linear decay
Warmup epochs	3.0
Momentum	0.937
Weight decay	0.0005
Maximum epochs	300
Early stopping patience	50
Batch size	16
Image resolution	640 × 640
Data augmentation	Mixed (see Section 4.2)

4. Results

This section presents the experimental evaluation of the proposed method. For clarity and completeness, we first describe the experimental settings, including the hardware/software environment and datasets, followed by quantitative comparisons, ablation studies, and cross-dataset experiments.

4.1. Experimental environment

The experimental platform consists of an Intel Core i7-14700K processor (20 cores, 28 threads, 3.4 GHz), 64 GB of Random Access Memory (RAM), and an NVIDIA GeForce RTX 4090D Graphics Processing Unit (GPU) with 24 GB of video memory. The software environment is built on Windows 10, using PyTorch 2.3.0 and Compute Unified Device Architecture (CUDA) 11.8 for model training and evaluation.

For models with framework-specific optimizations, such as PP-YOLOE implemented in PaddlePaddle, the corresponding native environments were configured to ensure fair and optimized performance comparisons.

The training hyperparameters are summarized in Table 2.

To mitigate stochastic variability during training, the reported quantitative results for all models correspond to the mean performance over five independent runs with different random seeds. This protocol ensures statistically stable and reproducible comparisons by reducing the influence of initialization randomness.

4.2. Datasets

To evaluate the performance and generalization capability of the proposed AGE-YOLO model for both steel surface inspection and cross-domain defect detection, three publicly available datasets are employed: GC10-DET (Zhang, 2019), NEU-DET (Park, 2022), and HRIPCB (Huang and Wei, 2019). Among them, GC10-DET is selected as the primary benchmark due to its high-resolution imagery, complex industrial backgrounds, and the presence of low-contrast and visually ambiguous defects, which pose greater challenges to robust detection. NEU-DET and HRIPCB are further introduced to evaluate cross-dataset generalization under standardized steel surface and PCB inspection scenarios, respectively.

- **GC10-DET dataset** (Global Cold-rolled Steel Surface Defect Dataset, Fig. 6): Collected from real cold-rolled steel production lines, GC10-DET is selected as our primary benchmark due to its unconstrained imaging conditions and process-specific complexities. It comprises ten representative defect categories: punching holes (ph), weld line (wl), crescent gap (cg), water spot (ws), oil spot (os), silk spot (ss), inclusion (in), rolled pit (rp), crease (cr), and waist folding (wf). The dataset contains 2294 grayscale images with a high resolution of 2048 × 1000 pixels. A significant challenge in this dataset is the presence of low-contrast defects, such as water and oil spots, which exhibit high inter-class similarity and are prone to false positives. We split the data using an 8:1:1 ratio, applying a ×4 augmentation to the training set.
- **NEU-DET dataset** (Northeastern University Steel Surface Defect Dataset): Released by Northeastern University, NEU-DET is a widely used benchmark for steel surface defect detection. It includes six defect types — crazing (cr), inclusion (in), patches (pa), pitted surface (ps), rolled-in scale (rs), and scratches (sc) — with 300 grayscale images per class, totaling 1800 images at 200 × 200 pixels. We split the data using an 8:1:1 ratio, applying a ×5 augmentation to the training set. Despite its smaller size and simpler backgrounds compared with GC10-DET, NEU-DET's standardized annotations make it suitable for cross-dataset validation.
- **HRIPCB dataset** (Printed Circuit Board Defect Dataset): Developed by the Human–Robot Interaction (HRI) Laboratory at Peking University, this dataset is widely used for high-resolution electronics inspection. It contains six defect categories: missing hole (mh), mouse bite (mb), open circuit (oc), short (sh), spur (sp), and spurious copper (sc). The official release comprises 1386 images, including original captures and their rotated variants. To prevent potential data leakage caused by sample redundancy, the dataset was partitioned based on the 653 unique base images, which were divided into training, validation, and test sets using an approximate 6:2:2 ratio. This split was designed to allocate a relatively larger and more diverse test set, thereby improving the statistical reliability of evaluation on this compact dataset. The training set was subsequently expanded to 750 samples by incorporating the provided rotated images. A ×6 augmentation strategy was then applied to the training data, while the validation and test sets remained composed solely of independent original images to ensure a fair and unbiased assessment.

To enhance generalization and alleviate overfitting, data augmentation was applied to the training sets. The augmentation pipeline includes random horizontal flipping ($p = 0.5$), random scaling within the range $[0.8, 1.2]$, and photometric transformations. In particular, HSV jittering was applied to the saturation and value channels with a factor of 0.2, corresponding to an illumination variation range of $[0.8, 1.2]$. This configuration reflects the controlled yet variable lighting conditions typical of industrial inspection environments, thereby improving robustness to localized shadows and reflections without introducing unrealistic visual artifacts.

Additionally, Mosaic augmentation was employed during training to merge four samples into a single composite image, further increasing background diversity and improving the representation of small-scale defects.

In summary, the chosen datasets enable a hierarchical evaluation protocol. GC10-DET is adopted as the primary benchmark due to its high-resolution imagery and complex industrial backgrounds (e.g., oil spots and low-contrast water patterns), which pose greater challenges than the relatively standardized samples in NEU-DET. This choice aligns with recent studies (Yeung and Lam, 2022; Liu et al., 2024), highlighting that the scale variability and morphological diversity of GC10-DET are critical for assessing the practical deployability of modern detection frameworks. After establishing performance on GC10-DET, we further validate the model's robustness on NEU-DET and HRIPCB using independent training-from-scratch settings (without pre-training or fine-tuning), thereby providing a systematic and unbiased evaluation of the model's generalization capability.

4.3. Evaluation metrics

To comprehensively evaluate the detection performance, we adopt several standard metrics widely used in object detection, including Precision (P), Recall (R), Average Precision (AP), and mean Average Precision (mAP).

Precision reflects the reliability of the predicted positive samples and is defined as:

$$P = \frac{TP}{TP + FP}, \tag{23}$$

where TP and FP denote the numbers of true positives and false positives, respectively.

Recall measures the capability of the model to identify all ground-truth objects and is computed as:

$$R = \frac{TP}{TP + FN}, \tag{24}$$

where FN represents the number of false negatives.

Average Precision (AP) summarizes the detection performance for a single class by integrating the Precision–Recall curve:

$$AP = \int_0^1 P(R)\, dR. \tag{25}$$

For multi-class detection tasks, mean Average Precision (mAP) is used to aggregate performance across all categories:

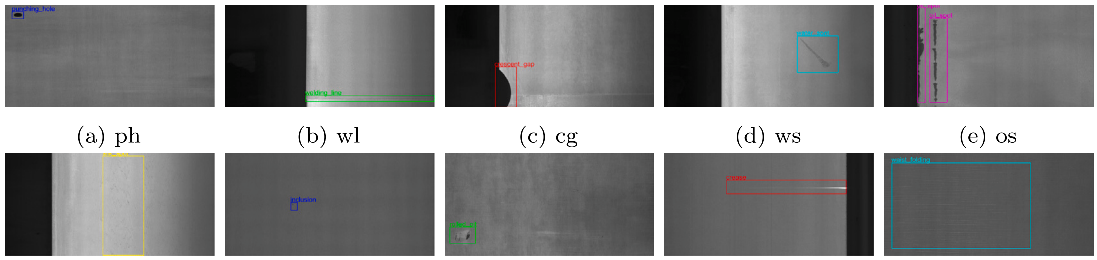
$$mAP = \frac{1}{C} \sum_{i=1}^{C} AP_i, \tag{26}$$

where C denotes the total number of classes and AP_i corresponds to the AP of the ith class.

Beyond detection accuracy, model efficiency is also considered. Specifically, the number of parameters (in millions, M) and computational complexity (in Giga Floating-Point Operations, G) are reported to reflect model compactness and computational cost. Inference efficiency is further evaluated using frames per second (FPS). All metrics and units are consistently applied throughout the comparative, ablation, and cross-dataset experiments to ensure fair evaluation.

4.4. Comparative experiments

The performance of AGE-YOLO, built upon YOLOv11n, was evaluated on the GC10-DET dataset. Comparisons were conducted against the classical two-stage detector Faster R-CNN and multiple one-stage YOLO detectors, including YOLOv3-tiny, YOLOv5n, YOLOv8n, and the YOLOv11 series (n, s, m, and l). In addition, more recent lightweight variants such as YOLOv12n and YOLOv13n were included. Transformer-based detectors, namely Real-Time DEtection TRansformer (RT-DETR) and Differentiable Fine-grained Information



(a) ph (b) wl (c) cg (d) ws (e) os

(f) ss (g) in (h) rp (i) cr (j) wf

Fig. 6. Examples of the ten defect categories in the GC10-DET dataset.

Network (DFINE-n), were also incorporated to enable a comprehensive comparison across both convolutional and Transformer paradigms. Furthermore, efficiency-oriented benchmarks such as NanoDet-plus (including its light version) and PP-YOLOE-tiny were evaluated to provide a thorough comparison with ultra-lightweight and evolved convolutional architectures. YOLOv11n served as the baseline for quantifying the performance gains achieved by AGE-YOLO.

Most models were trained using the hyperparameters summarized in Table 2, with consistent data preprocessing and augmentation settings. For a fair comparison, the input resolution was fixed at 640 × 640 for all detectors. Exceptions include RT-DETR, which employed a smaller batch size (8) due to GPU memory constraints, and DFINE-n, which followed its official training configuration (160 epochs, AdamW optimizer, batch size 8). The detailed quantitative results are summarized in Table 3.

As summarized in Table 3, AGE-YOLO achieves the highest mAP@0.5 of 72.8%, surpassing the YOLOv11n baseline (67.6%) by 5.2 percentage points. Based on the five-run evaluation protocol, AGE-YOLO exhibits a small standard deviation of ±0.23%, indicating stable improvements across different random initializations. While the parameter count and computational cost increase slightly (3.41M and 7.7G, respectively), the model remains within the ultra-lightweight regime, maintaining a superior precision-to-parameter ratio compared with standard YOLO scaling (e.g., YOLOv11s and YOLOv11m).

The per-class analysis demonstrates that AGE-YOLO effectively addresses the multifaceted challenges of industrial inspection, including small-scale anomalies, irregular morphologies, and low-contrast background interference. Specifically, for rolled pit (rp) and inclusion (in), categories characterized by tiny spatial footprints and visually ambiguous boundaries, the model achieves substantial improvements, with rp increasing from 16.1% to 28.5%. This highlights the capability of SIFHead to capture fine-grained localization cues for small targets. Furthermore, AGE-YOLO shows strong performance on non-structured defects such as silk spot (ss, 57.8% to 63.1%), where the gated linear modulation in FC-MANet effectively suppresses high-frequency textural noise and emphasizes irregular defect contours. Notably, for categories that tend to blend into the background, such as water spot (ws), the mAP increases significantly from 79.5% to 91.4%. This 11.9% gain confirms that the ACF module successfully recalibrates feature responses, preventing subtle low-contrast signals from being overwhelmed by complex industrial backgrounds.

Compared with efficiency-oriented models, AGE-YOLO demonstrates a clear advantage in balancing compactness and representational capacity. While NanoDet-plus (64.1%) and PP-YOLOE-tiny (62.5%) provide strong efficiency, their representational capability is insufficient for the complex backgrounds present in GC10-DET. AGE-YOLO surpasses NanoDet-plus by 8.7 percentage points, indicating that the proposed mechanisms effectively bridge the gap between extreme lightweight design and high detection accuracy.

Within the YOLO series, YOLOv11m achieves a competitive mAP (70.5%); however, its computational cost reaches 67.7G FLOPs, which is approximately 8.8× higher than that of AGE-YOLO (7.7G). This increased complexity is accompanied by a notable reduction in inference speed, with FPS decreasing from 255 to 117. Furthermore, recent variants such as YOLOv12n and YOLOv13n maintain compactness but exhibit plateaued accuracy (around 67%), suggesting that excessive lightweighting may limit feature expressiveness. AGE-YOLO alleviates this limitation by emphasizing architectural efficiency rather than simple model scaling. In comparison with Transformer-based detectors, AGE-YOLO outperforms the strongest baseline, DFINE-n (69.4%), by 3.4 percentage points while remaining more parameter-efficient and over twice as fast (189 FPS vs. 88 FPS), making it better suited for high-throughput industrial deployment.

Beyond deep learning approaches, AGE-YOLO significantly outperforms traditional machine vision methods, which typically achieve an mAP of 43%–47% on similar metal surface benchmarks (Ma et al., 2024; Usamentiaga et al., 2022). Our framework therefore provides an efficient end-to-end solution that overcomes the representation limitations of hand-crafted features.

For a more intuitive trade-off analysis, Fig. 7 illustrates the mAP–FPS distribution, where the bubble size represents the parameter count (M). AGE-YOLO occupies the top-most position region of the Pareto front, indicating a favorable balance between detection accuracy and real-time inference speed under moderate model complexity.

To evaluate the model's interpretability and robustness, Gradient-weighted Class Activation Mapping (Grad-CAM) visualizations were generated for three representative defect types: inclusion, rolled pit, and crease (Fig. 8). As illustrated in the "Ours" column of Fig. 8, AGE-YOLO exhibits highly localized activation responses that demonstrate superior alignment with annotated defect boundaries compared to the baselines.

Specifically, for inclusion (in), which are often small and multi-instance, AGE-YOLO isolates each anomaly with sharp focal points, whereas baselines like YOLOv11l show broader, more diffuse activation patterns. In the case of rolled pit (rp), the model effectively filters out high-amplitude textural noise from the steel surface, concentrating exclusively on defect-sensitive regions. For elongated non-structured defects such as crease (cr), AGE-YOLO maintains a continuous and stable response distribution along the defect's longitudinal contour, avoiding the fragmented activations seen in rigid fusion schemes. Overall, these visualizations confirm that the integrated architectural enhancements in AGE-YOLO achieve a precise attention mechanism that remains concentrated on salient defect features while significantly reducing background interference.

In summary, the comparative results demonstrate that AGE-YOLO achieves a superior trade-off between detection precision and inference throughput compared to both convolutional and Transformer-based paradigms. By harmonizing architectural efficiency with representational power, the proposed framework fulfills the stringent requirements for real-time industrial surface inspection on resource-constrained edge platforms.

Table 3
Comparison of detection performance on the GC10-DET dataset. Each metric for mAP@0.5 is reported as mean ± standard deviation across five independent runs.

Model	mAP@0.5 (%)	Precision (%)	Recall (%)	ph (%)	wl (%)	cg (%)	ws (%)	os (%)	ss (%)	in (%)	rp (%)	cr (%)	wf (%)	Params (M)	FLOPs (G)	FPS[a]
Faster R-CNN (Ren et al., 2017)	57.5 ± 0.35	55.6	60.3	96.4	33.5	93.2	69.7	54.4	54.7	30.7	14.7	34.1	93.8	41.40	137.2	36
YOLOv3-tiny (Redmon and Farhadi, 2018)	60.9 ± 0.28	60.8	58.7	97.2	57.6	93.1	80.1	61.5	57.6	28.6	3.9	38.9	90.8	12.10	19.1	487
YOLOv5n (Jocher, 2020)	66.1 ± 0.14	75.6	62.3	98.2	90.0	92.3	76.5	66.8	60.3	35.1	13.8	34.9	93.0	2.51	7.2	265
YOLOv8n (Jocher et al., 2023)	66.7 ± 0.15	68.2	65.3	98.6	86.5	97.4	82.5	61.9	56.3	39.7	7.2	38.8	98.3	3.01	8.2	296
YOLOv11n (Khanam and Hussain, 2024)	67.6 ± 0.11	64.3	67.6	97.2	93.9	91.4	79.5	73.4	57.8	38.7	16.1	32.6	95.3	2.58	6.3	255
YOLOv11s (Khanam and Hussain, 2024)	68.8 ± 0.13	73.1	67.7	98.6	88.1	92.1	81.6	74.9	60.2	44.1	17.5	33.4	97.9	9.41	21.3	232
YOLOv11m (Khanam and Hussain, 2024)	70.5 ± 0.12	76.2	66.1	99.1	91.5	93.2	87.4	74.4	62.7	40.4	18.2	41.9	95.7	20.04	67.7	117
YOLOv11l (Khanam and Hussain, 2024)	68.3 ± 0.16	75.1	64.6	99.0	93.4	94.5	87.0	71.4	53.9	40.7	12.1	38.9	92.4	25.30	87.3	108
YOLOv12n (Tian et al., 2025)	67.0 ± 0.15	68.8	63.9	97.0	91.5	90.3	85.1	67.1	59.0	32.0	13.4	37.3	97.7	2.50	5.8	159
YOLOv13n (Lei et al., 2025)	67.2 ± 0.14	65.8	67.2	96.9	88.2	96.8	81.7	72.7	59.5	35.9	13.3	31.8	94.8	2.46	6.3	118
RT-DETR (Zhao et al., 2024)	69.2 ± 0.28	78.8	68.1	99.6	96.7	95.3	79.3	78.3	54.5	44.9	8.7	35.6	99.5	32.80	108.0	40
DFINE-n (Peng et al., 2025)	69.4 ± 0.26	72.6	64.7	99.1	95.7	94.2	83.6	71.9	57.9	41.3	17.6	36.8	96.2	3.70	7.1	88
NanoDet-plus-light (Lyu, 2021)	60.5 ± 0.18	62.4	58.2	94.3	74.6	88.9	72.3	65.0	53.5	34.0	10.2	28.7	83.8	0.49	1.6	178
NanoDet-plus (Lyu, 2021)	64.1 ± 0.20	65.5	60.9	95.1	76.0	95.0	81.1	72.5	55.1	36.8	13.8	31.5	84.4	1.17	3.6	138
PP-YOLOE-tiny (Xu et al., 2022)	62.5 ± 0.24	64.1	61.7	93.9	70.4	93.9	71.9	68.2	52.5	37.1	14.8	34.8	87.2	4.42	9.8	98
Ours (AGE-YOLO)	**72.8** ± **0.23**	72.1	68.4	95.7	89.6	93.6	91.4	77.1	63.1	47.6	28.5	44.2	97.2	3.41	7.7	189

Note: M = Millions, G = Giga FLOPs (Floating Point Operations), FPS = Frames Per Second. PP-YOLOE was evaluated using its native PaddleDetection framework. All reported metrics represent the mean of five independent runs. For defect abbreviations, see Section 4.2.

[a] Testing environment: NVIDIA RTX 4090D (24 GB), batch size = 1, resolution = 640 × 640. FPS denotes the end-to-end inference speed, encompassing data preprocessing, model forward pass, and NMS post-processing.

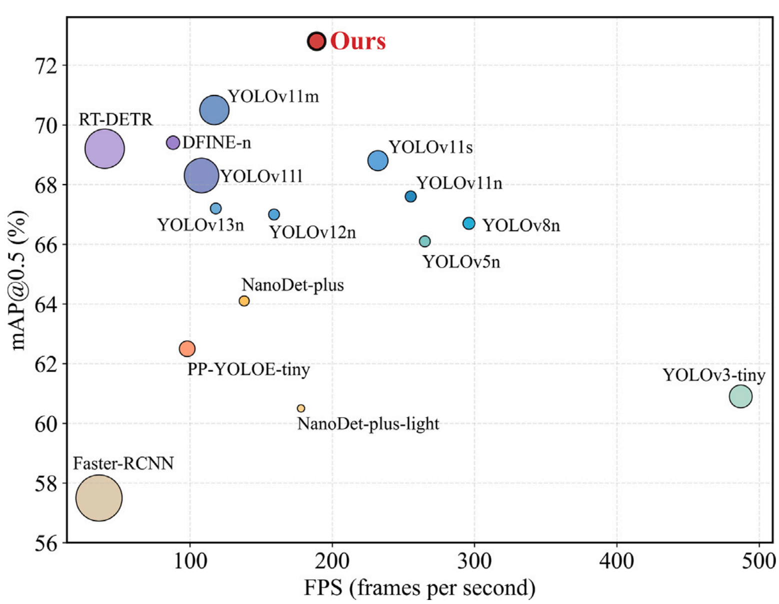


Fig. 7. Accuracy–speed comparison among different detection models. The horizontal axis represents inference speed (FPS), the vertical axis shows mAP@0.5 (%), and the circle size denotes the number of parameters (M). AGE-YOLO achieves a favorable trade-off between detection accuracy and real-time performance, combining high precision with fast inference.

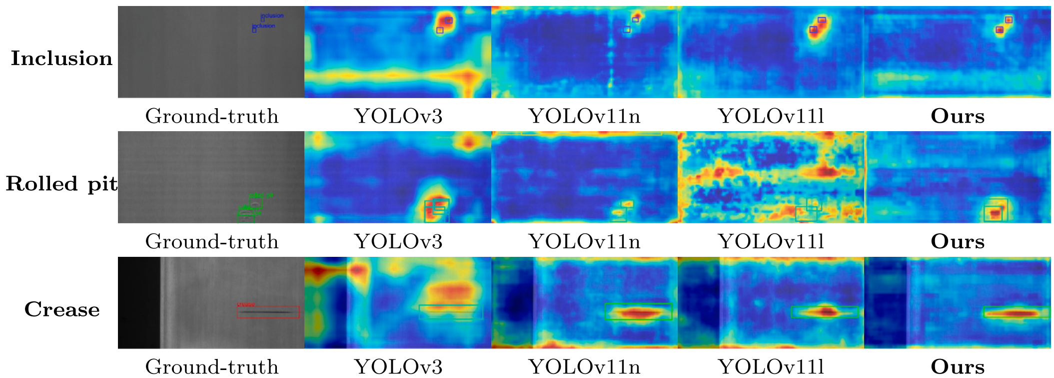


Fig. 8. Grad-CAM visualization of representative defect categories (inclusion, rolled pit, and crease). Each row corresponds to a defect category, and columns denote different model configurations. Red regions represent strong activations around defect areas, while blue regions indicate weak or background responses. AGE-YOLO shows enhanced localization and attention consistency compared with baseline models.

Table 4
Module-level ablation study on the GC10-DET dataset. Each sub-section investigates the internal components of FC-MANet, ACF, and SIFHead, respectively. All metrics represent the mean values across five independent runs.

Model	mAP@0.5 (%)	Precision (%)	Recall (%)	ph (%)	wl (%)	cg (%)	ws (%)	os (%)	ss (%)	in (%)	rp (%)	cr (%)	wf (%)	Params (M)	FLOPs (G)
FC-MANet module ablation															
Baseline	67.6	64.3	67.6	97.2	93.9	91.4	79.5	73.4	57.8	38.7	16.1	32.6	95.3	2.58	6.3
w/o Faster-CGLU	68.4	68.2	66.5	97.9	91.8	93.2	81.5	74.0	58.6	39.5	18.2	34.1	95.2	2.70	6.5
w/o PartialConv	69.5	73.1	65.1	98.0	90.8	94.8	82.1	74.8	60.2	41.2	20.5	36.4	96.2	3.00	7.1
FC-MANet (Full)	**70.6**	74.8	65.2	98.1	91.5	95.2	83.2	74.5	61.5	42.1	22.8	38.4	98.7	3.08	7.2
ACF module ablation															
Baseline	67.6	64.3	67.6	97.2	93.9	91.4	79.5	73.4	57.8	38.7	16.1	32.6	95.3	2.58	6.3
w/o SE (Channel)	68.8	68.4	67.2	97.8	93.6	92.0	82.5	75.1	57.5	39.1	17.8	35.4	97.2	3.10	7.8
w/o PA (Spatial)	69.3	70.2	67.9	97.5	93.8	92.2	83.1	74.5	57.8	39.4	18.5	35.8	96.4	3.10	7.8
ACF (Full)	**70.2**	71.2	68.5	97.8	94.2	92.5	84.6	76.8	58.6	39.5	19.4	42.1	96.5	3.29	8.2
SIFHead module ablation															
Baseline	67.6	64.3	67.6	97.2	93.9	91.4	79.5	73.4	57.8	38.7	16.1	32.6	95.3	2.58	6.3
w/o Shared Conv	68.7	66.8	68.2	97.3	91.5	92.4	80.8	73.9	57.9	41.2	17.5	36.4	98.1	2.85	7.0
BN instead of GN	69.2	70.4	66.8	97.4	87.2	94.2	86.8	73.2	58.4	43.6	15.6	36.8	98.8	2.20	4.9
w/o SimAM	69.1	68.2	68.9	97.0	92.1	92.8	81.6	74.2	58.1	43.5	19.1	37.5	95.1	2.20	4.9
SIFHead (Full)	**69.8**	69.5	69.4	97.2	92.8	93.1	82.5	74.8	58.2	45.1	20.4	38.6	95.3	2.20	4.9

4.5. Ablation experiments

To examine the effectiveness of the proposed FC-MANet, ACF, and SIFHead modules, a series of ablation experiments were conducted on the GC10-DET dataset, with YOLOv11n used as the baseline. All ablation settings follow the same training protocol to ensure fair comparison. The evaluation is performed at both the overall performance level and the defect-class level.

4.5.1. Module-level ablation study

The module-level ablation study investigates the individual contributions of the three proposed components and their internal submodules. For each module, the complete design serves as the reference, while ablated variants are constructed by removing or substituting specific elements. All models were evaluated under a consistent protocol of five independent runs to ensure statistical reliability. The quantitative results are summarized in Table 4, and the corresponding qualitative visualizations are presented in Fig. 9.

The results for FC-MANet indicate that its integration facilitates a significant precision gain, rising from 64.3% to 74.8% (a 10.5% absolute increase). This empirical evidence suggests that the combination of Faster-CGLU and Partial Convolution effectively functions as a discriminative feature filter that suppresses background textural noise. As shown in the first row of Fig. 9, the baseline exhibits diffuse activations for the silk spot (ss) category, whereas the full FC-MANet configuration produces a more concentrated response that aligns with the irregular defect contours. The gated linear modulation ensures that non-structural cues are preserved without being diluted by high-frequency surface patterns, resulting in a 3.7% accuracy improvement for this category.

For the ACF module, the ablation data reveals its primary role in mitigating cross-scale misalignment and background interference. The inclusion of ACF leads to consistent improvements in water spot (ws, +5.1%) and oil spot (os, +3.4%) detection, which are categories characterized by low visual contrast. The second row of Fig. 9 illustrates that while ablated variants often produce fragmented activations for elongated defects like creases (cr), the full ACF module restores structural continuity via joint channel-spatial attention. This confirms that the proposed normalization and attention-based fusion strategies maintain numerical stability and semantic consistency, preventing subtle defect signals from being overwhelmed by complex industrial backgrounds during feature integration.

The SIFHead module achieves a superior balance between computational efficiency and localization precision. Notably, it reduces the parameter count from 2.58M to 2.20M (a 14.7% reduction) while improving the overall mAP. The most significant categorical gain is observed in the crease (cr) category, which increases from 32.6% to 38.6%. This improvement suggests that the shared convolutional encoding and SimAM-based 3D energy recalibration provide a more stable focus for small-scale anomalies. Furthermore, as indicated in Table 4, substituting Group Normalization (GN) with Batch Normalization (BN) leads to a degradation in detection stability, validating the

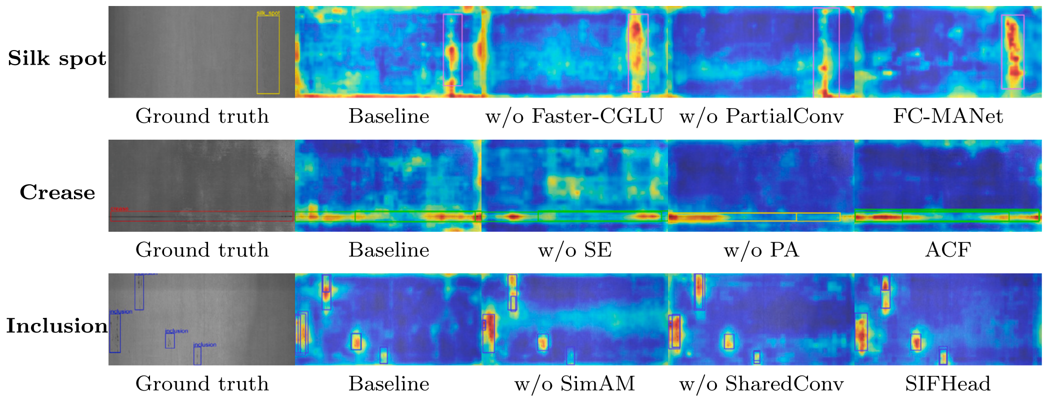


Fig. 9. Visualization of Grad-CAM heatmaps for the module-level ablation study. The rows display representative defect categories: silk spot, crease, and inclusion. The columns, from left to right, represent the ground truth image, the baseline (YOLOv11n), variants without specific sub-components, and the final proposed modules (FC-MANet, ACF, and SIFHead). The visualization suggests that the proposed components encourage more spatially focused and semantically discriminative activations on the actual defect regions.

Table 5
Mapping of observed feature sensitivities to AGE-YOLO components.

Component	Observed sensitivity	Design rrationale
FC-MANet	Baseline exhibits diffuse activation and background leakage on faint textures (e.g., silk spots).	Utilizes gated linear modulation to act as a high-frequency filter, isolating defect-related signals from stochastic surface noise.
ACF	Baseline shows fragmented or disconnected activations for elongated, continuous defects (e.g., creases).	Employs dual-attention recalibration to restore structural topology, ensuring semantic continuity during the fusion of heterogeneous scale information.
SIFHead	Baseline fails to maintain a pinpointed focus on small anomalies (e.g., inclusions) across varying scales.	Enforces scale-invariant encoding via weight sharing and SimAM energy recalibration, stabilizing localization while reducing parameter redundancy by 14.7%.

importance of batch-independent normalization for consistent feature statistics. The third row of Fig. 9 corroborates these findings, showing that SIFHead encourages more semantically aligned activation maps for small inclusions (in) compared to the baseline.

In summary, the module-level ablation results demonstrate that FC-MANet, ACF, and SIFHead contribute to the AGE-YOLO framework through complementary mechanisms. By refining local feature responses, ensuring cross-scale structural continuity, and stabilizing localization at the prediction head, these modules collectively facilitate a more precise and interpretable detection process for diverse industrial surface defects.

4.5.2. Linking feature sensitivities to architectural design

To provide a concrete rationale for the architectural choices in AGE-YOLO, this section establishes a structural mapping between defect morphologies and the corresponding modules validated in the ablation study. This alignment ensures that the observed feature sensitivities in the Grad-CAM analysis are directly addressed by specific design components, as summarized in Table 5.

The synergy among these components transforms the detection process from a rigid feature-extraction pipeline into a morphology-aware attention mechanism. For example, the transition from fragmented to continuous activations observed in the second row of Fig. 9 indicates that the ACF module not only amplifies signal responses but also reconstructs the topological representation of defects. Likewise, the reduction in background dispersion achieved by FC-MANet further validates its role in enhancing discriminative capability without resorting to excessive parameter growth. This consistent alignment between visual evidence and architectural design provides a data-driven explanation for the performance improvements observed in the primary benchmarks.

4.5.3. Combination-level ablation study

To investigate the synergistic interactions between the proposed components, a combination-level ablation study was conducted. This evaluation transitions from single-module integrations to dual-module and full-module configurations, with quantitative results summarized in Table 6.

The results indicate that incorporating any single module improves the baseline mAP@0.5, with each component exhibiting distinct performance characteristics. FC-MANet provides a substantial precision enhancement (64.3% to 74.8%), albeit with a slight reduction in recall, suggesting a more discriminative feature filtering process. Conversely, ACF and SIFHead focus on improving recall and localization stability, particularly for categories involving complex backgrounds or multi-scale features.

The integration of two modules leads to performance gains that exceed individual enhancements, indicating complementary interactions. The combination of FC-MANet and ACF achieves 72.0% mAP@0.5, demonstrating that refined local features and adaptive multi-scale fusion collectively improve robustness against low-contrast and irregular defects. A noteworthy observation occurs when SIFHead is integrated with the FC+ACF configuration: the full AGE-YOLO framework achieves the highest accuracy of 72.8% while simultaneously reducing the parameter count from 3.79M to 3.41M. This indicates that the shared convolutional design and 3D energy recalibration in SIFHead not only enhance representation power but also optimize architectural efficiency.

The category-wise analysis in Table 6 further confirms this complementarity. While FC-MANet is particularly effective for non-structured defects like silk spots (ss), the inclusion of ACF and SIFHead is essential for stabilizing the detection of elongated creases (cr) and small inclusions (in). This suggests that the proposed modules do not simply target isolated stages of the network but form a coherent framework

Table 6
Ablation study of different module combinations on the GC10-DET dataset. The combination of FC-MANet, ACF, and SIFHead yields the best overall accuracy, demonstrating the complementary effect of the three modules.

Model	mAP@0.5 (%)	P (%)	R (%)	ph	wl	cg	ws	os	ss	in	rp	cr	wf	Params (M)	FLOPs (G)
Baseline	67.6	64.3	67.6	97.2	93.9	91.4	79.5	73.4	57.8	38.7	16.1	32.6	95.3	2.58	6.3
+FC-MANet (FC)	70.6	74.8	65.2	98.1	91.5	95.2	83.2	74.5	61.5	42.1	22.8	38.4	98.7	3.08	7.2
+ACF	70.2	71.2	68.5	97.8	94.2	92.5	84.6	76.8	58.6	39.5	19.4	42.1	96.5	3.29	8.2
+SIFHead	69.8	69.5	69.4	97.2	92.8	93.1	82.5	74.8	58.2	45.1	20.4	38.6	95.3	2.20	4.9
+FC+ACF	72.0	74.2	67.3	98.4	92.8	95.8	86.2	77.5	62.4	43.8	24.6	38.8	99.9	3.79	9.1
+FC+SIFHead	71.8	73.2	68.8	97.8	91.2	96.2	88.5	75.8	62.8	46.5	25.1	40.2	93.9	2.70	5.8
+ACF+SIFHead	71.4	70.8	70.2	98.1	92.5	94.6	88.2	76.4	60.2	45.8	22.4	42.8	93.0	2.91	6.8
AGE-YOLO	72.8	72.1	68.4	95.7	89.6	93.6	91.4	77.1	63.1	47.6	28.5	44.2	97.2	3.41	7.7

Table 7
Impact of data augmentation strategies on detection performance (mAP@0.5) on the GC10-DET dataset.

Setting	Flip	Scale	HSV	Mosaic	mAP@0.5 (%)
Baseline	×	×	×	×	71.3
+ Geo	✓	✓	×	×	71.6
+ Photo	✓	✓	✓	×	72.1
Full	✓	✓	✓	✓	**72.8**

where feature refinement, adaptive fusion, and stable prediction heads work in tandem to address the diverse challenges of industrial surface inspection.

4.5.4. Effectiveness of data augmentation

To quantitatively assess the impact of the adopted augmentation pipeline, a stepwise ablation study was performed on the GC10-DET dataset. All augmentation configurations strictly adhere to the parameter settings described in Section 4.2. The corresponding results are presented in Table 7.

Starting from the baseline without augmentation, the introduction of geometric transformations (random horizontal flipping and multi-scale scaling) increases the mAP@0.5 from 71.3% to 71.6%, indicating improved robustness to spatial variations and scale diversity. Incorporating photometric perturbations via HSV jittering further raises performance to 72.1%, reflecting enhanced tolerance to illumination changes under realistic industrial inspection conditions. Finally, adding Mosaic augmentation leads to the best performance of 72.8%, suggesting that enriched contextual diversity and increased exposure to small defects positively influence detection accuracy.

Overall, the cumulative gain of +1.5% over the non-augmented baseline demonstrates that the designed augmentation strategy improves model generalization and robustness in a controlled and physically consistent manner.

4.6. Cross-dataset validation experiments

To comprehensively evaluate the generalization capability of the proposed method under diverse industrial scenarios, cross-dataset validation experiments are conducted on the NEU-DET and HRIPCB datasets. To ensure a rigorous and unbiased assessment of architectural robustness, all models in this section are trained from scratch on their respective datasets. No pre-trained weights from GC10-DET or external sources are utilized, and no cross-dataset fine-tuning or domain adaptation is performed.

The evaluated models include YOLOv5n, YOLOv11n/l, RT-DETR, DFINE-n, NanoDet-plus, and the proposed AGE-YOLO. Quantitative results on NEU-DET and HRIPCB are reported in Tables 8 and 9, respectively. On NEU-DET, AGE-YOLO achieves an mAP@0.5 of 85.4%, outperforming the YOLOv11n baseline by 4.1 percentage points. Notably, on the crazing (cr) category — which features complex and fine-grained textures — AGE-YOLO maintains a significant lead over NanoDet-plus (+14.7% AP). This indicates that while extreme lightweighting in NanoDet-plus limits representational capacity for subtle anomalies, the gated modulation in FC-MANet effectively preserves these critical features even in constrained environments.

On the HRIPCB dataset, AGE-YOLO achieves an mAP@0.5 of 95.7%. Although its accuracy is slightly lower than that of the large-scale RT-DETR (96.3%), AGE-YOLO requires only 10.4% of the parameters (3.41M vs. 32.8M), demonstrating markedly higher efficiency for high-precision electronics inspection. The successful transfer from metallic steel surfaces (characterized by specular reflections) to PCB substrates (composed of structured composite materials) highlights the model's strong cross-material generalization capability. This cross-domain performance can be primarily attributed to the adaptive feature modeling strategy rather than reliance on domain-specific priors. Specifically, the ACF module dynamically balances multi-scale features through learnable cross-scale compensation, effectively mitigating scale distribution discrepancies between steel and PCB defects. In addition, SIFHead promotes scale-invariant feature encoding via shared convolutions while stabilizing feature statistics with Group Normalization (GN), thereby reducing sensitivity to domain-dependent background variations. Finally, FC-MANet further enhances robustness by strengthening the representation of irregular and low-contrast defect textures common across diverse industrial scenarios. Collectively, these complementary mechanisms enable AGE-YOLO to capture domain-invariant defect characteristics without requiring cross-dataset fine-tuning.

Qualitative results in Fig. 10 further illustrate this robust behavior. As shown in the visualizations for NEU-DET (Fig. 10(a)), AGE-YOLO accurately localizes fine-grained crazing and elongated patches where baselines often exhibit missed detections (marked by red boxes). Similarly, for the HRIPCB dataset (Fig. 10(b)), the framework demonstrates high sensitivity to subtle defects like missing hole and open circuit, maintaining tighter bounding box alignment compared to larger architectures. These findings confirm that AGE-YOLO captures universal industrial defect signatures rather than overfitting to domain-specific priors.

Despite these promising results, several boundary conditions remain. Extremely low-contrast defects under strong background textures continue to pose challenges, as their feature responses may be partially suppressed. In addition, environmental disturbances such as motion blur caused by mechanical vibrations or severe occlusions can adversely affect detection precision. Addressing these edge cases through more robust context-aware modulation or dedicated temporal augmentation strategies constitutes an important direction for future work.

Overall, the cross-dataset validation results demonstrate that AGE-YOLO maintains stable detection performance across diverse defect distributions and imaging conditions. Under comparable computational budgets, the proposed method exhibits competitive performance relative to baseline models, indicating enhanced robustness and generalization for cross-domain industrial defect detection.

4.7. Practical deployment and edge-side performance

To assess the practical deployability of AGE-YOLO, performance benchmarks were carried out on an NVIDIA Jetson Orin Nano (8 GB) embedded platform. This device, delivering up to 40 Tera Operations Per Second (TOPS) of artificial intelligence computing capability,

Table 8
Performance comparison of different models on the NEU-DET dataset.

Model	mAP@0.5 (%)	Precision (%)	Recall (%)	cr (%)	in (%)	pa (%)	ps (%)	rs (%)	sc (%)	Params (M)	FLOPs (G)
YOLOv5n (Jocher, 2020)	79.2 ± 0.21	76.9	72.9	55.7	89.5	92.9	82.2	61.6	93.3	2.51	7.2
YOLOv11n (Khanam and Hussain, 2024)	81.3 ± 0.14	74.3	75.9	58.6	92.1	91.0	84.4	67.2	94.5	2.58	6.3
YOLOv11l (Khanam and Hussain, 2024)	79.9 ± 0.25	82.7	68.9	52.4	90.0	93.4	81.8	68.7	92.9	25.30	87.3
RT-DETR (Zhao et al., 2024)	76.8 ± 0.32	83.0	64.5	52.4	90.3	91.5	72.4	60.7	93.5	32.80	108.0
DFINE-n (Peng et al., 2025)	86.6 ± 0.19	86.8	81.9	74.7	91.4	93.0	87.8	83.3	89.3	3.70	7.1
NanoDet-plus (Lyu, 2021)	77.1 ± 0.24	76.2	73.6	50.6	89.3	86.0	80.8	62.9	93.0	1.17	3.6
Ours (AGE-YOLO)	85.4 ± 0.18	83.0	79.0	65.3	92.5	92.7	92.3	75.1	94.2	3.41	7.7

Table 9
Performance comparison of different models on the HRIPCB dataset.

Model	mAP@0.5 (%)	Precision (%)	Recall (%)	mh (%)	mb (%)	oc (%)	sh (%)	sp (%)	sc (%)	Params (M)	FLOPs (G)
YOLOv5n (Jocher, 2020)	90.5 ± 0.15	92.2	85.7	98.0	84.3	93.9	91.5	86.1	89.3	2.51	7.2
YOLOv11n (Khanam and Hussain, 2024)	90.8 ± 0.11	93.2	82.4	99.1	84.4	96.2	90.1	85.5	89.4	2.58	6.3
YOLOv11l (Khanam and Hussain, 2024)	94.1 ± 0.18	93.2	90.4	99.3	90.9	97.3	94.1	91.9	91.3	25.30	87.3
RT-DETR (Zhao et al., 2024)	96.3 ± 0.20	96.8	93.7	99.1	97.7	99.4	91.8	93.3	96.4	32.80	108.0
DFINE-n (Peng et al., 2025)	91.6 ± 0.17	92.7	87.2	97.6	89.4	94.6	90.2	87.5	90.4	3.70	7.1
NanoDet-plus (Lyu, 2021)	87.9 ± 0.22	88.2	85.3	92.4	87.5	89.4	88.0	83.6	86.5	1.17	3.6
Ours (AGE-YOLO)	95.7 ± 0.12	94.4	93.3	99.4	96.5	99.2	95.7	92.2	91.2	3.41	7.7

represents a typical edge-computing terminal in modern smart manufacturing environments. To maximize inference efficiency, the NVIDIA TensorRT engine was employed with FP16 quantization and operator fusion enabled.

The experimental results show that for a 640 × 640 input, the end-to-end latency, including preprocessing, model inference, and Non-Maximum Suppression (NMS) post-processing, averages 13.4 ms, corresponding to a sustained throughput of 74.6 Frames Per Second (FPS). Given that many steel inspection lines operate at approximately 30 FPS, AGE-YOLO provides a latency headroom of nearly 20 ms per frame. This computational margin is essential for coordinating downstream industrial processes such as Programmable Logic Controller (PLC) communication, input/output (I/O) control, and real-time defect logging.

Moreover, the architectural efficiency of AGE-YOLO, driven by the parameter-shared SIFHead and adaptive feature fusion within the ACF module, ensures that high detection accuracy is preserved without exceeding the memory and thermal constraints of the embedded platform. The observed computational margin therefore supports seamless integration into existing Automated Optical Inspection (AOI) systems, enabling a practical transition from laboratory validation to pilot deployment on production lines.

5. Conclusion

This work presents AGE-YOLO, a lightweight and deployment-oriented detection framework for industrial surface defect inspection. To address critical challenges, including multi-scale feature complexity, irregular defect morphology, data imbalance, and stringent real-time constraints, AGE-YOLO incorporates three complementary modules. FC-MANet enhances nonlinear feature representation and inter-channel interaction, enabling more effective modeling of irregular defect patterns. ACF performs adaptive cross-scale feature fusion via joint spatial–channel guidance, thereby improving robustness against complex industrial backgrounds. SIFHead reinforces the localization of small and fine-grained defects by integrating lightweight shared convolutions with scale-adaptive normalization under limited computational budgets. Extensive experiments on the GC10-DET, NEU-DET, and HRIPCB datasets demonstrate that AGE-YOLO achieves competitive performance among existing YOLO-based and Transformer-based approaches, striking a favorable balance among detection accuracy, inference efficiency, and cross-dataset generalization. In particular, it attains a peak accuracy of 72.8% mAP@0.5 on GC10-DET, representing a 5.2% improvement over the YOLOv11 baseline. Moreover, the high inference speed observed on edge hardware (approximately 74.6 FPS) further confirms its practicality for integration into NVIDIA Jetson-powered automated sorting systems and PLC-controlled defect logging units.

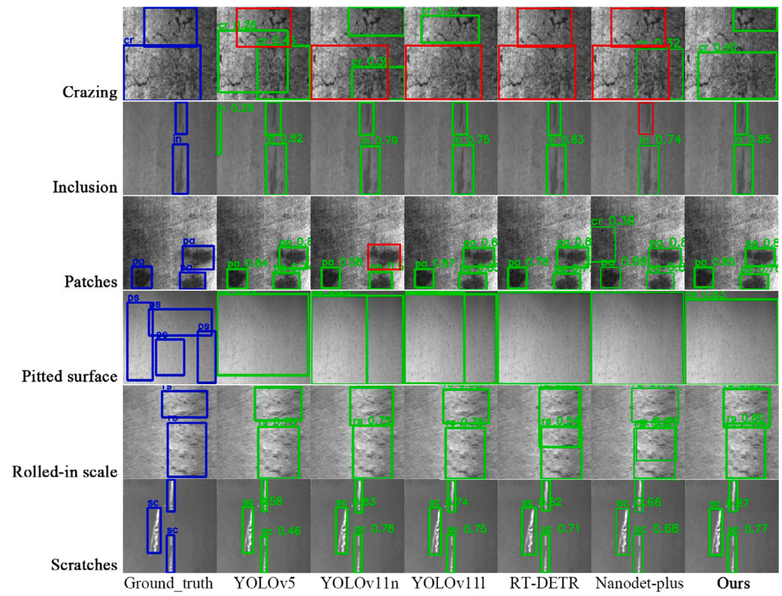
Regarding scalability, we plan to extend the framework to a broader range of industrial materials, such as aluminum alloys, textiles, and optical films, where complex interactions between illumination and material texture may introduce additional challenges for feature alignment. Although performance fluctuations are expected due to variations in optical reflectivity and stochastic surface textures, the successful validation on both metallic (steel) and composite (PCB) substrates establishes a solid foundation for cross-material generalization. Building upon the demonstrated edge-side feasibility, future work will incorporate knowledge distillation techniques to further compress the model for deployment in highly resource-constrained environments. To improve robustness under complex real-world conditions, we will also explore few-shot learning and domain adaptation strategies to address challenges associated with limited labeled data, severe illumination inconsistencies, and motion blur caused by mechanical vibrations. In addition, active learning mechanisms will be developed to enable continual adaptation to evolving defect patterns in dynamic production settings. Collectively, these efforts aim to ensure that the proposed system remains reliable, efficient, and precise across diverse and demanding industrial scenarios.

CRediT authorship contribution statement

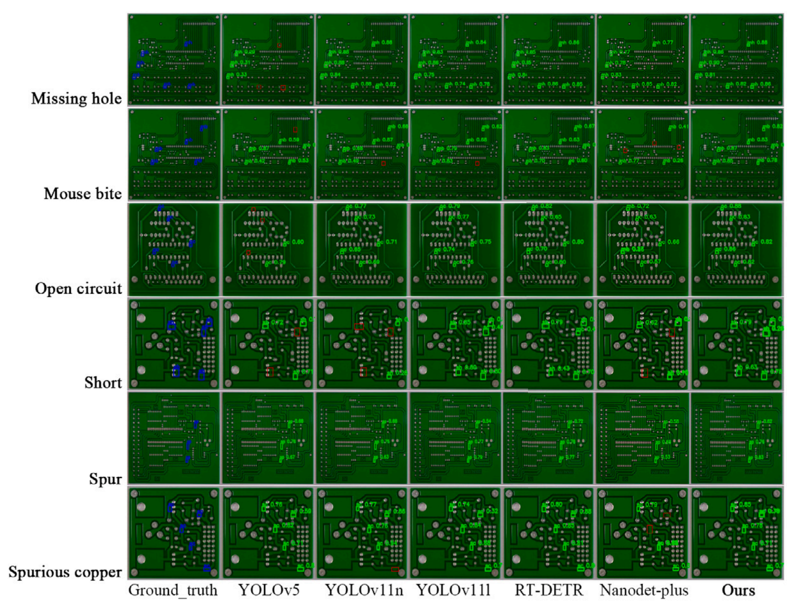
Hao Yan: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Lanxiang Chen:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Hong Zhang:** Writing – review & editing, Validation, Methodology. **Shikun Chen:** Writing – review & editing, Validation, Methodology, Conceptualization. **Shunwu Xu:** Writing – review & editing, Validation, Software. **Zhaowen Chen:** Writing – review & editing, Validation, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



(a) NEU-DET



(b) HRIPCB

Fig. 10. Qualitative results of cross-dataset detection on NEU-DET and HRIPCB datasets. Blue boxes denote ground truth annotations, green boxes indicate predicted detections, and red boxes highlight missed detections. AGE-YOLO shows fewer false or missed detections and maintains consistent accuracy across domains.

Data availability

Data will be made available on request.

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